Study of the effect of correlated experimental uncertainty on Bayesian Inference of quark–gluon plasma properties

MSc. student: Nara Avila Moraes

Advisor: **Prof. Dr. Matthew William Luzum**

Instituto de Física / USP

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Summary

1. Tasks Status

Motivation

Constrains dos parametros do QGP estao cada vez mais precisos, recentemente utilizando analise bayesiana.
Inspiração: "Inspiração: Perhaps most notably, I report the most precise estimate to date of the temperature-dependent specific shear viscosity η/s, the measurement of which is a primary goal of heavy-ion physics. The estimated minimum value is $\eta/s = 0.085^{+0.026}_{-0.025}$ (posterior median and 90% uncertainty), remarkably close to the conjectured lower bound of $1/4\pi \simeq 0.08$." [1]
Continuacao: Then, for any given parameter values, we compute the likelihood by evaluating the model and calculating the fit to data, folding in any sources of uncertainty from the model calculation and experimental measurements. Finally, we invoke Bayes' theorem, which states that the posterior is proportional to the product of the likelihood and the prior: [1])
Encadeamento de incertezas no processo (desenhar se possivel): Armed with a dataset, a computational model, and Bayes' theorem, we can compute the posterior probability at any point in parameter space. To extract estimates of each parameter, we must now construct their marginal distributions, obtained for any given parameter by marginalizing over

References I

[1] Jonah E. Bernhard. "Bayesian parameter estimation for relativistic heavy-ion collisions". en. arXiv:1804.06469 [nucl-th]. PhD thesis. arXiv, Apr. 2018. URL: http://arxiv.org/abs/1804.06469 (visited on 12/12/2025).